2ª Lista de Exercícios de Cálculo I

1) Calcule $f'(p)$ pela definição, sendo dados

$a)\ y = \text{sen } 4x$

$b)\ y = \cos 5x$

$c)\ f(x) = e^{3x}$

$d)\ f(x) = \cos 8x$

$e)\ y = \text{sen } t^3$

$f)\ g(t) = \ln(2t + 1)$

$g)\ x = e^{\text{sen } t}$

$h)\ f(x) = \cos e^x$

$i)\ y = (\text{sen } x + \cos x)^3$

$j)\quad y = \sqrt{3x + 1}$

$l)\quad f(x) = \sqrt[3]{\dfrac{x - 1}{x + 1}}$

$m)\ y = e^{-5x}$

$n)\ x = \ln(t^2 + 3t + 9)$

$o)\ f(x) = e^{\text{tg } x}$

$p)\ y = \text{sen}(\cos x)$

$q)\ g(t) = (t^2 + 3)^4$

$r)\ f(x) = \cos(x^2 + 3)$

$s)\quad y = \sqrt{x + e^x}$

$t)\ y = \text{tg } 3x$

$u)\ y = \sec 3x$

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2) Calcule a 2ª derivada

a) $y = \text{sen } 5t$

b) $y = \cos 4t$

c) $x = \text{sen } \omega t$, ω constante

d) $y = e^{-3x}$

e) $y = e^{-x^2}$

f) $y = \dfrac{e^x}{x+1}$

g) $y = \ln(x^2 + 1)$

h) $y = \dfrac{x^2}{x-1}$

i) $y = e^{-x} - e^{-2x}$

j) $y = e^{-x} \cos 2x$

l) $y = \dfrac{x}{x^2+1}$

m) $y = \dfrac{3x+1}{x^2+x}$

n) $y = \dfrac{\text{sen } 3x}{e^x}$

o) $y = xe^{-2x}$

p) $y = \text{sen}(\cos x)$

q) $f(x) = \dfrac{4x+5}{x^2-1}$

r) $y = xe^{\frac{1}{x}}$

s) $y = \dfrac{x^2}{x^2+x+1}$

t) $g(t) = \sqrt{t^2+3}$

u) $y = x\sqrt[3]{x+2}$

3) Seja $g: \mathbb{R} \to \mathbb{R}$ tal que $g(1) = 2$ e $g'(1) = 3$. Calcule $f(0)$, sendo f dada por $f(x) = e^x g(3x+1)$.

4) Seja $f: \mathbb{R} \to \mathbb{R}$ derivável até segunda ordem e seja g dada por $g(x) = f(e^{2x})$. Verifique que

$$g''(x) = 4e^{2x}[f'(e^{2x}) + e^{2x} f''(e^{2x})]$$

5) Calcule a derivada

a) $f(x) = 5^x + \log_3 x$

b) $y = 2^{x^2} + 3^{2x}$

c) $g(x) = 3^{2x+1} + \log_2 (x^2 + 1)$

d) $y = (2x + 1)^x$

e) $f(x) = x^{\text{sen } 3x}$

f) $g(x) = (3 + \cos x)^x$

g) $y = x^x \text{ sen } x$

h) $y = x^{x^2+1}$

i) $y = (1 + i)^{-t}$, i constante

j) $y = 10^x - 10^{-x}$

l) $y = (2 + \text{sen } x)^{\cos 3x}$

m) $y = \ln (1 + x^x)$

6) Determine a função que é dada implicitamente pela equação

$$xy^2 + x + y = 1$$

7) A função $y = f(x)$ é dada implicitamente pela equação $xy + 3 = 2x$.

Mostre que $x\frac{dy}{dx} = 2 - y$ e calcule $\left.\frac{dy}{dx}\right|_{x=2}$.

8) Expresse $\frac{dy}{dx}$ em termos de x e y, sabendo $y = f(x)$ é uma função diferenciável dada pela equação

a) $x^2 - y^2 = 4$

b) $y^3 + x^2 y = x + 4$

c) $xy^2 + 2y = 3$

d) $y^5 + y = x$

e) $x^2 + 4y^2 = 3$

f) $xy + y^3 = x$

g) $x^2 + y^2 + 2y = 0$

h) $x^2 y^3 + xy = 2$

i) $xe^y + xy = 3$

j) $y + \ln(x^2 + y^2) = 4$

l) $5y + \cos y = xy$

m) $2y + \text{sen } y = x$

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SOLUÇÃO

Exercício 1

a) $4\cos 4x$ *b)* $-5\,\text{sen}\,5x$ *c)* $3e^{3x}$ *d)* $-8\,\text{sen}\,8x$

e) $3t^2\cos t^3$ *f)* $\dfrac{2}{2t+1}$ *g)* $e^{\text{sen}\,t}\cos t$ *h)* $-e^x\,\text{sen}\,e^x$

i) $3(\text{sen}\,x+\cos x)^2(\cos x-\text{sen}\,x)$ *j)* $\dfrac{3}{2\sqrt{3x+1}}$

l) $\dfrac{2}{3(x+1)^2}\sqrt[3]{\left(\dfrac{x+1}{x-1}\right)^2}$ *m)* $-5e^{-5x}$ *n)* $\dfrac{2t+3}{t^2+3t+9}$ *o)* $e^{\text{tg}\,x}\sec^2 x$

p) $-\text{sen}\,x\cos(\cos x)$ *q)* $8t(t^2+3)^3$ *r)* $-2x\,\text{sen}(x^2+3)$

s) $\dfrac{1+e^x}{2\sqrt{x+e^x}}$ *t)* $3\sec^2 3x$ *u)* $3\sec 3x\,\text{tg}\,3x$

Exercício 2:

a) $-25\,\text{sen}\,5t$ *b)* $-16\cos 4t$ *c)* $-w^2\,\text{sen}\,wt$ *d)* $9e^{-3x}$

e) $2e^{-x^2}(2x^2-1)$ *f)* $\dfrac{e^x(x^2+1)}{(x+1)^3}$ *g)* $\dfrac{2(1-x^2)}{(x^2+1)^2}$ *h)* $\dfrac{2}{(x-1)^3}$

i) $e^{-x}-4e^{-2x}$ *j)* $e^{-x}(4\,\text{sen}\,2x-3\cos 2x)$ *l)* $\dfrac{2x(x^2-3)}{(x^2+1)^3}$

m) $\dfrac{2(3x^3+3x^2+3x+1)}{(x^2+x)^3}$ *n)* $\dfrac{-2[4\,\text{sen}\,3x+3\cos 3x]}{e^x}$

o) $4e^{-2x}(x-1)$ *p)* $-\cos x\cos(\cos x)-\text{sen}^2 x\,\text{sen}(\cos x)$

q) $\dfrac{8x^3+30x^2+24x+10}{(x^2-1)^3}$ *r)* $\dfrac{e^{1/x}}{x^3}$ *s)* $\dfrac{2(-x^3-3x^2+1)}{(x^2+x+1)^3}$

t) $\dfrac{3}{(t^2+3)\sqrt{t^2+3}}$ *u)* $\dfrac{4x+12}{9\sqrt[3]{(x+2)^5}}$

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Exercício 3: 8

Exercício 5

a) $5^x \ln 5 + \dfrac{1}{x \ln 3}$ *b)* $2x\,2^{x^2} \ln 2 + 2 \cdot 3^{2x} \ln 3$

c) $2 \cdot 3^{2x+1} \ln 3 + \dfrac{2x}{(x^2+1)\ln 2}$ *d)* $(2x+1)^x \left[\ln(2x+1) + \dfrac{2x}{2x+1}\right]$

e) $x^{\operatorname{sen} 3x}\left[3\cos 3x \ln + \dfrac{\operatorname{sen} 3x}{x}\right]$ *f)* $(3+\cos x)^x \left[\ln(3+\cos x) - \dfrac{x \operatorname{sen} x}{3+\cos x}\right]$

g) $x^x[(1+\ln x)\operatorname{sen} x + \cos x]$ *h)* $x^{x^2+1}\left[2x \ln x + \dfrac{x^2+1}{x}\right]$

i) $-(1+i)^{-t} \ln(1+i)$ *j)* $(10^x + 10^{-x}) \ln 10$

l) $(2+\operatorname{sen} x)^{\cos 3x}\left[-3\operatorname{sen} 3x \ln(2+\operatorname{sen} x) \dfrac{\cos x \cos 3x}{2+\operatorname{sen} x}\right]$

m) $\dfrac{x^x(1+\ln x)}{1+x^x}$

Exercício 6: $y = \dfrac{-1+\sqrt{-4x^2+4x+1}}{2x}$

Exercício 7: $\dfrac{3}{4}$

Exercício 8:

a) $\dfrac{dy}{dx} = \dfrac{x}{y}$ *b)* $\dfrac{dy}{dx} = -\dfrac{2xy-1}{3y^2+x^2}$ *c)* $\dfrac{dy}{dx} = -\dfrac{y^2}{2xy+2}$

d) $\dfrac{dy}{dx} = \dfrac{1}{1+5y^4}$ *e)* $\dfrac{dy}{dx} = -\dfrac{x}{4y}$ *f)* $\dfrac{dy}{dx} = \dfrac{1-y}{x+3y^2}$

g) $\dfrac{dy}{dx} = -\dfrac{x}{y+1}$ *h)* $\dfrac{dy}{dx} = -\dfrac{2xy^3+y}{3x^2y^2+x}$ *i)* $\dfrac{dy}{dx} = -\dfrac{y+e^y}{xe^y+x}$

j) $\dfrac{dy}{dx} = -\dfrac{2x}{x^2+y^2+2y}$ *l)* $\dfrac{dy}{dx} = \dfrac{y}{5-\operatorname{sen} y - x}$ *m)* $\dfrac{dy}{dx} = \dfrac{1}{2+\cos y}$